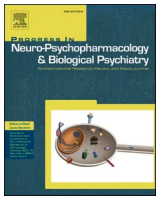
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Screens and brains: multimodal neuroimaging insights into mechanisms of problematic smartphone use

Nadine D. Wolf [a,b], Gudrun M. Henemann [a,b], Mike M. Schmitgen [a,b], Julian Koenig [c], Patrick Bach [b,d], Robert C. Wolf [a,b,*]

[a] Department of General Psychiatry, Center for Psychosocial Medicine, Heidelberg University, Germany
[b] DZPG, German Center for Mental Health, partner site Mannheim/Heidelberg/Ulm (ZIHUb), Germany
[c] University of Cologne, Faculty of Medicine and University Hospital Cologne, Department of Child and Adolescent Psychiatry, Psychosomatics and Psychotherapy, Cologne, Germany
[d] Department of Addictive Behavior and Addiction Medicine, Central Institute of Mental Health, Mannheim, Germany

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ABSTRACT

Background: Problematic smartphone use (PSU) is increasingly regarded as a form of behavioral addiction, marked by compulsive engagement and diminished self-regulation. An expanding body of evidence points to distinct neural correlates of PSU, including heightened sensitivity to smartphone-related cues, aberrant reward processing, and executive dysfunction.
Objectives: This systematic review synthesizes contemporary evidence from multimodal neuroimaging studies that seek to elucidate the neurobiological substrates of PSU. To unify and contextualize these findings, a social-cognitive framework is proposed.
Methods: A pre-registered systematic literature review (PROSPERO ID no. 1068662) was conducted to identify structural and functional neuroimaging studies on problematic smartphone use (PSU) published in the past decade.
Results: Neuroimaging data consistently demonstrate abnormalities in frontostriatal and frontolimbic circuits, implicating reduced executive control, heightened cue reactivity, and altered processing of socially salient stimuli. Neural responses to smartphone-related reward cues exhibit substantial convergence with activation patterns associated with social cognition. Social-cognitive distortions, such as maladaptive interpretations of digital social feedback, may potentiate the motivational salience of smartphone cues, thereby reinforcing compulsive usage patterns.
Conclusion: Reviewed studies indicate that PSU emerges from a dynamic and multifactorial interaction between neurobiological vulnerabilities, affective responsiveness, cognitive biases, and impairments in executive control. These factors show patterns of reciprocal reinforcement, facilitating the consolidation of automatic, habitual engagement with smartphone-mediated stimuli. Within this framework, socially mediated reinforcement mechanisms may serve as critical drivers for both, the development and persistence of PSU. Integration of social-cognitive factors of PSU offers promising avenues for future prevention and intervention strategies, highlighting targets for future investigations.

1. Introduction

The widespread adoption of smartphones has profoundly transformed modern life. By 2023, approximately 4.24 billion people worldwide were using a smartphone, with projections for 2024 approaching 4.87 billion. In Germany, smartphone use among adolescents has consistently exceeded 90 % for several years (https://de.statista.com/, 2025). Far beyond serving as mere communication tools, smartphones have become ever-present companions integral to social interaction, information retrieval, and digital entertainment. Simultaneously, mounting evidence suggests that high levels of smartphone engagement are associated with a range of negative outcomes, including reduced academic and occupational performance, diminished quality of life, cognitive impairments, and increased risk of sleep disturbances,

* Corresponding author at: Center for Psychosocial Medicine, Department of General Psychiatry, Heidelberg University, Voßstraße 4, 69115 Heidelberg, Germany.
E-mail address: christian.wolf@med.uni-heidelberg.de (R.C. Wolf).

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anxiety, and depression (De-Sola Gutierrez et al., 2016; Sohn et al., 2019; Mac, 2020). At the same time, an increasing body of evidence indicates that elevated levels of smartphone engagement can be linked to various adverse outcomes, such as reduced academic and occupational performance, diminished quality of life, cognitive difficulties, and heightened risk of sleep disturbances, anxiety, and depression. However, it is important to note that such associations are not uniform across all populations; in certain age groups (Chen et al., 2024) or specific academic contexts (Kertechian and Ismail, 2025), higher levels of smartphone use may also be associated with beneficial effects.

These observed behavioral patterns have occurred alongside evolving terminology in the field. Early in the digital era, expressions such as *mobile-phone addiction* were common; however, around 2010, the term *smartphone addiction* (SPA) gained traction, reflecting conceptual proximity to behavioral addictions such as Internet Gaming Disorder (IGD), as defined in the DSM-5 appendix (Zhang et al., 2024). More recently, however, the emphasis has shifted away from the device itself. Increasingly, researchers prefer the terms *excessive* or *problematic smartphone use* (PSU), emphasizing not the smartphone as a medium, but the nature and context of its use (Panova and Carbonell, 2018; Montag et al., 2019a; James et al., 2023). This terminological refinement supports a dimensional view of PSU (Kwako et al., 2018; Brand et al., 2025), ranging from unproblematic use to maladaptive patterns and, in some cases, addiction-like constellations (Wolf and Wolf, 2016). Although PSU is not formally recognized as a disorder, parallels can be drawn to the DSM-5 criteria for substance use disorders and IGD, as well as the ICD-11 definition of gaming disorder, converging on **impaired control over the behavior, continued use despite negative consequences**, and **significant functional impairment** in daily life. Conceptualizing PSU along this spectrum allows for a balanced perspective without pathologizing normative use.

Alongside conceptual developments, neuroscientific interest in PSU has grown markedly. Recent research has shifted from focusing on the investigation of isolated neural correlates to broader cognitive processes that underpin compulsive behaviors. At present, PSU is thought to reflect a dysbalance between automatic, reward-driven impulses and top-down regulatory control. Highly salient stimuli, e.g. push notifications, likes, or incoming messages, tend to capture attention rapidly and disproportionately, while the capacity to inhibit immediate urges in favor of long-term goals appears diminished (Schmitgen et al., 2020; Wolf and Wolf, 2020). The attentional bias toward digitally mediated rewards is further associated with reduced cognitive flexibility and impaired executive function (Henemann et al., 2023a; Lin et al., 2022).

Against the backdrop of an expanding body of neuroimaging research, the present review pursues two central aims: first, to systematically synthesize current findings from structural and functional neuroimaging studies related to PSU; second, to interpret these findings within the framework of the I-PACE model (Interaction of Person – Affect – Cognition – Execution), which conceptualizes behavioral addictions as the result of complex interactions among individual predispositions, affective dysregulation, cognitive biases, and deficits in executive control (Brand et al., 2019). More recent refinements of the model emphasize the roles of gratification and compensation, highlighting both positive and negative reinforcement, as well as conditioning processes, habit formation, and specific situational cues (Brand et al., 2025). Building on this framework, the presented review explores the relevance of social-cognitive processes for the development and maintenance of PSU. In this context, we highlight the role of reward networks in social reinforcement processes (Zaff et al., 2023), particularly the pursuit of social belonging and the avoidance of exclusion (Tomova et al., 2021; Wesselmann and Parris, 2021). This integrative approach aims to provide a more comprehensive understanding of how neural, psychological, and social mechanisms converge to promote and sustain problematic patterns of smartphone use. In doing so, this review seeks to inform the development of targeted prevention and intervention strategies that address both neurocognitive and socio-affective dimensions of PSU.

2. Methods

This systematic review was conducted in adherence to the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines, see also Fig. 1. Following preregistration in PROSPERO (ID no. 1068662) a literature search was performed using the PubMed database, covering publications from January 2015 to April 2025. The search was restricted to peer-reviewed articles published in English and focused on neuroimaging studies examining PSU. The search strategy involved querying titles and abstracts for studies that included the term "smartphone" in conjunction with descriptors such as "excessive," "problematic," "addiction," or "dependence," combined with neuroimaging-related terms including "MRI," "fMRI," "imaging," "neuroimaging," or broader terms such as "brain," as well as compound expressions linking "brain" with either "structure," "function," "activation," or "activity". Studies were eligible for inclusion if they met the following criteria: (1) investigated problematic or excessive smartphone use or smartphone addiction, (2) employed structural or functional neuroimaging methodologies, and (3) were published in English. Exclusion criteria comprised: (1) review articles and meta-analyses, (2) publications providing only an abstract without full-text access, and (3) studies with incomplete or insufficient data. Initial screening, data extraction, and cross-verification and final inclusion decisions were made jointly by two authors (NDW and RCW). Extracted data encompassed the following variables: author(s), year of publication, participant demographics (mean age and gender), psychometric instruments used to assess PSU, the neuroimaging modality applied, and, in case of functional imaging, the experimental tasks.

3. Results

A total of 35 studies met the inclusion criteria, including 9 studies utilizing structural imaging techniques and 24 employing functional imaging methods. Two studies used both structural and functional neuroimaging.

Across the identified structural neuroimaging studies, publication years ranged from 2016 to 2023. Sample sizes varied considerably, from 19 to 266 individuals, with 8 studies reporting PSU and control groups separately and 3 studies reporting combined samples with no specific information with regard to subgroup sample size, sex or gender. While some studies achieved a relatively balanced sex distribution, several leaned toward female-majority samples. Mean participant ages spanned from early adolescence (~15 years) to young adulthood (~23 years) across both PSU and control groups. Regarding psychometric scales addressing PSU, ~45 % of studies employed the Smartphone Addiction Scale (SAS-SV (Kwon et al., 2013)), ~27 % the Smartphone Addiction Proneness Scale (SAPS (Kim et al., 2014)), while the remaining studies relied on other scales.

The functional neuroimaging studies identified were published between 2017 and 2025. Sample sizes varied widely, from 16 to 162 individuals, with 18 studies reporting PSU and control groups separately and 7 studies reporting age and sex data from combined samples. Several studies achieved a relatively balanced sex distribution, while others included clearly unbalanced cohorts. The mean age of participants spanned from early adolescence (~13 years) to adulthood (~30 years). Regarding PSU scales, 50 % of studies employed the SAS-SV, ~30 % the SAPS, while the remaining studies used other psychometric instruments. In terms of methodology, the majority of studies (75 %) used fMRI, whereas 25 % relied on other modalities including EEG, fNIRS, and MR-spectroscopy. Within the fMRI subset, ~44 % employed resting-state imaging, while the remaining fMRI studies employed various experimental tasks. Study details are presented in Supplementary Tables 1 and 2, arranged chronologically, with structural studies listed first, followed by functional investigations.

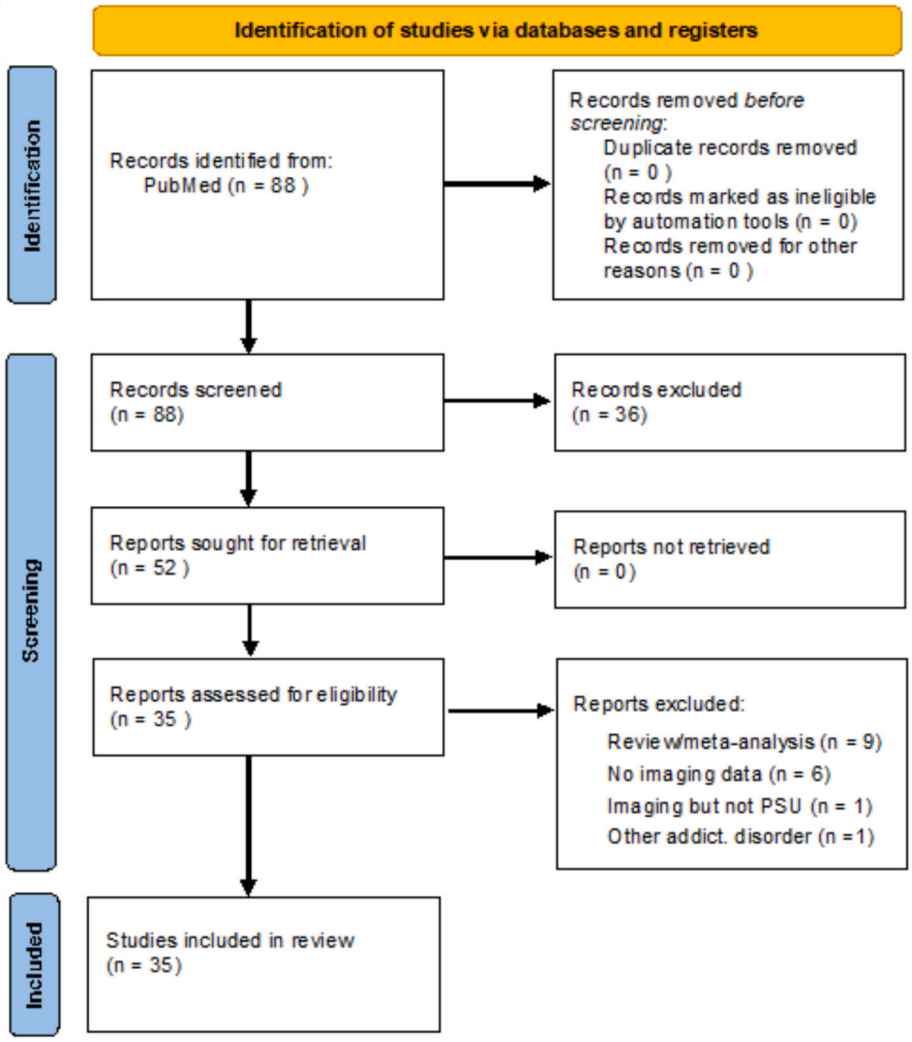


Fig. 1. PRISMA flow diagram of the systematic review.

The following sections present findings from structural neuroimaging first, followed by functional imaging data. The latter are subdivided into results from resting-state paradigms, capturing intrinsic neural activity (INA), and findings derived from task-based functional imaging approaches. Additionally, a distinction is made between cross-sectional and longitudinal study designs.

3.1. Structural imaging

Structural neuroimaging studies have revealed a consistent pattern of anatomical alterations in individuals with PSU. Reduced gray matter volume (GMV) has been observed in the left anterior insular cortex, the inferior temporal gyrus, and the parahippocampal cortex. Moreover, higher scores on the Smartphone Addiction Inventory (SPAI) (Lin et al., 2014) have been negatively associated with GMV in both the anterior cingulate cortex (ACC) and the orbitofrontal cortex (OFC) (Horvath et al., 2020). Notably, the finding of reduced OFC volume and its inverse correlation with PSU severity, as assessed via the Smartphone Addiction Proneness Scale (SAPS) (Kim et al., 2014), has been independently replicated (Lee et al., 2019). The association between ACC volume and PSU has likewise been corroborated in further studies (Zou et al., 2021). Studies of cortical surface morphology point to altered gyral configuration, with a reduction in cortical folding complexity in the right superior frontal gyrus and the ACC among individuals with PSU (Hirjak et al., 2022). Of further interest is a recent investigation of cortical surface markers that identified a significant relationship between cortical thickness in the precuneus, PSU severity, and levels of Fear of Missing Out (FoMO (Wang et al., 2023);). FoMO refers to the pervasive apprehension that one might miss socially significant experiences or information. Indeed, the association between FoMO and problematic digital media use is well documented and supported by meta-analytic evidence (Zhang et al., 2023). These findings suggest that altered precuneus morphology may reflect a neural substrate for socio-affective vulnerabilities that contribute to compulsive smartphone engagement. Beyond cortical regions, additional evidence points to the involvement of subcortical structures in PSU, such as cerebellar and striatal regions (Yoo et al., 2021; Cho et al., 2021). Volume reductions have been reported in the left caudate nucleus, with significant associations between caudate volume, SAPS scores, and serum glutamate levels. A mediation analysis further revealed that glutamate concentration exerted an indirect effect on PSU severity via caudate volume, suggesting specific neurochemical contributions to structural vulnerability (Yoo et al., 2021). Complementing these volumetric findings, diffusion tensor imaging (DTI) studies have demonstrated alterations in white matter integrity (WMI). Specifically, PSU has been associated with microstructural changes in interhemispheric fiber tracts as well as in frontostriatal and limbic pathways, including connections involving the amygdala and hippocampus (Zou et al., 2021; Hu et al., 2017; Tymofiyeva et al., 2020; Wang et al., 2016).

3.2. Resting-state imaging

INA, as examined through resting-state imaging, refers to spontaneous brain activity occurring in the absence of external stimuli or task demands. It reflects a fundamental property of functional brain organization, enabling continuous communication between distributed neural systems. As such, studies of INA provide crucial insights into the stability and integrity of large-scale functional networks, independent of extrinsic task-demand (Rosazza and Minati, 2011).

Several studies have investigated INA in the context of PSU. Convergent with structural studies, aberrant INA in the ACC in individuals with PSU compared to control participants has been observed, and neural activity in this region has been shown to negatively correlate with the extent of smartphone use, as measured by the Smartphone Addiction Inventory (SPAI (Horvath et al., 2020)). Similar results have been reported for the insular cortex, with notable associations between insular/striatal and insular/OFC connectivity and prolonged smartphone use in bed (Paik et al., 2019). These findings align with evidence of disrupted connectivity between the striatum, particularly the nucleus accumbens, and the OFC, a pattern linked to withdrawal symptoms and cortisol release in PSU (Chun et al., 2018). Associations between prefrontal INA and cognitive control have also been reported, with significant associations between frontolimbic connectivity, self-control, and the duration of smartphone use (Pyeon et al., 2021). Yet, INA changes in PSU are not confined to isolated brain regions but likely reflect a broader dysfunction of large-scale neural networks: combined structural (GMV) and functional (INA) neuroimaging approaches using parallel Independent Component Analysis (pICA) have revealed anatomically overlapping GMV/INA systems. Two distinct neural networks exhibited characteristic alterations in individuals with PSU: a medial/dorsolateral prefrontal network showed reduced connectivity, while a parietal/corticocerebellar system demonstrated increased network coupling. The connectivity strength of these networks was significantly related to smartphone use duration and sleep disturbances (Schmitgen et al., 2022). Further contributions of executive, attentional, and salience-processing networks to PSU have been supported by three additional studies on intrinsic network connectivity (Afra et al., 2024; Kwon et al., 2022; Ahn et al., 2021). In particular, the attentional network has been linked to PSU severity (Kwon et al., 2022). Converging evidence also suggests that higher levels of smartphone use are associated with reduced temporal inflexibility of dynamic connectivity patterns within the attentional network (Liu et al., 2022). Expanding on contributions of executive and attention networks, a recent resting-state-EEG study suggested that alpha- and delta-band brain networks are significantly associated with the severity of PSU, and that negative emotions mediate this association within the alpha-band (Yin et al., 2024).

3.3. Task-based imaging

Task-based approaches complement resting-state investigations by capturing stimulus-evoked neural processes. Whereas INA-based imaging primarily reflects baseline, intrinsic network function, task-based paradigms allow for the targeted functional characterization of specific cognitive or affective processes under controlled experimental conditions. In the context of PSU, such paradigms have predominantly examined affective, attentional, and executive functions, as well as salience detection and reward processing.

3.3.1. Affective processing

Two fMRI focused on facial emotion recognition processing in PSU. In the context of the Facial Emotion Recognition Paradigm (Hariri et al., 2002), the experimental setting involved identifying **anger, fear, and sadness,** with neutral expressions used as comparisons. Significant positive associations were been reported between the extent of smartphone use and functional connectivity of the right amygdala with regions implicated in emotion regulation and self-referential processing. Specifically, individuals with PSU exhibited amygdala hyperconnectivity with the ACC and frontotemporal regions (Arato et al., 2023). Another study used a facial emotion discrimination task that tested both emotional valence (happy vs. angry faces) and emotional status (repetition vs. transition) (Lee et al., 2013). Compared to controls, persons with PSU showed reduced brain activity in several cortical and subcortical regions for both happy and angry faces, with more widespread reductions in the angry condition (e.g., precentral gyri, dorsolateral prefrontal cortex [DLPFC], dorsal ACC, and subcortical regions). For emotional status, individuals with PSU also showed reduced activity in multiple regions during both repetition and transition conditions, particularly in frontal, temporal, and parietal areas. Importantly, persons with PSU did not show greater activity than control participants in any condition, suggesting diminished cognitive control during multiple functional domains (valence and status) of affective processing (Chun et al., 2017).

3.3.2. Attention and executive function

Task-based investigations of attentional and executive processes have further delineated functional alterations associated with PSU. In a modified oddball paradigm incorporating distractor stimuli, individuals with PSU demonstrated reduced activation of the OFC (Han and Kim, 2022). Moreover, variations in frontal activation as a function of cognitive load were observed in both medial and lateral prefrontal regions. Under increasing task demands during a verbal interference paradigm (reading with distractors), individuals with PSU failed to exhibit the medial prefrontal activation increase typically seen in control participants and, additionally, showed a decline in lateral prefrontal activity (Asao et al., 2024). Increased distractibility in PSU was also observed alongside heightened activation within frontoparietal regions, suggesting reduced efficiency in the recruitment of attentional control networks (Choi et al., 2021). Complementing these observations, alterations in lateral prefrontal cortex activity, particularly within the dorsolateral region, have also been demonstrated across various experimental contexts (Liu et al., 2024; Liu et al., 2023; Li et al., 2023). Such aberrant prefrontal activation patterns in PSU are unlikely to reflect isolated regional deficits; rather, they appear indicative of broader executive network alterations. In line with this notion, a multimodal imaging study employing joint Independent Component Analysis (jICA) identified consistent activity patterns across several cognitive domains, including cue reactivity, inhibition, and working memory. This analysis revealed a process-independent frontoparietal network whose functional expression was significantly diminished in individuals with PSU relative to controls. Moreover, the connectivity strength within this network was significantly associated with daily smartphone use duration and craving (Henemann et al., 2023a).

3.3.3. Cue reactivity and reward processing

Cue reactivity (CR) is a well-established paradigm used to examine biobehavioral responses to addiction-relevant stimuli. Such cues, e.g., alcohol- or drug-related images in the context of substance use disorders, reliably engage reward and salience networks, including lateral prefrontal, cingulate, parietal, and precentral cortices, as well as the insula, striatum, and cerebellum (Starcke et al., 2018; Noori et al., 2016). In the context of PSU, this paradigm has been adapted by presenting smartphone-related stimuli, with a critical distinction made between inactive (screen-off) and active (screen-on) devices. In this regard, fMRI data revealed a complex pattern of regional hyper- and hypoactivation in individuals with PSU. Specifically, increased activation was observed in inferior prefrontal and occipital cortices, while reduced activity emerged in cingulate and bilateral temporoparietal regions (Fig. 2a). Notably, differential activation patterns were also found in additional salience-processing and motor-related areas, including the insular cortex, frontal operculum, ACC, and precentral gyrus, with activation levels varying depending on the stimulus condition (active vs. inactive smartphone cues). These neural responses were negatively associated

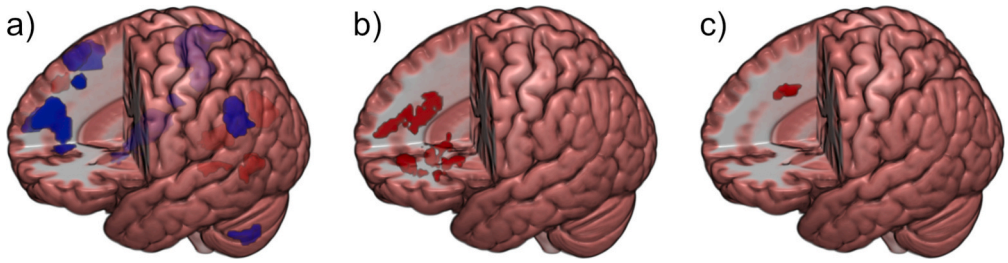


Fig. 2. Neural activity patterns associated with cue reactivity (a., b.) and social cognition (c.) in individuals with PSU compared to control participants.

a. Regions associated with cue reactivity showing hyperactivation and hypoactivation (depicted in red and blue, respectively) in individuals with PSU vs. controls. Image derived from ref. (Schmitgen et al., 2020).
b. CR-related hyperactivation following a three-day period of smartphone restriction (red). Image derived from ref. (Schmitgen et al., 2025).
c. Hyperactivation of the cingulate cortex in response to social exclusion in individuals with PSU compared to controls (red). Image derived from ref. (Henemann et al., 2025).

with PSU symptom dimensions such as compulsive use and withdrawal (Schmitgen et al., 2020). Further evidence linked CR-related neural activity in PSU to the endogenous opioid system, particularly the μ-opioid receptor (MOR) (Henemann et al., 2023b). Using a machine learning approach with several classification models, CR-associated activation patterns were shown to predict group membership (PSU vs. non-PSU) with an accuracy of approximately 80 % (Oweda et al., 2025). *Complementary EEG-research* showed that individuals with high levels of PSU exhibited larger reward positivity in response to social feedback compared to monetary feedback, unlike those with low PSU levels (Deng et al., 2021). Beyond CR-related responses, these findings suggest that heightened neural sensitivity to social rewards may play a role in the development or maintenance of PSU.

3.3.4. *Longitudinal studies*

In contrast to the predominance of cross-sectional studies reviewed thus far, longitudinal neuroimaging investigations in PSU remain markedly underrepresented: in a prospective study INA over a one-year follow-up period, baseline activity in the parahippocampal and medial temporal cortices was found to significantly moderate the relationship between PSU and depressive symptoms (Zou et al., 2022). In an interventional study, magnetic resonance spectroscopy revealed elevated GABA levels in ACC of individuals with PSU, which significantly declined following nine weeks of cognitive behavioral therapy (Seo et al., 2020). Longitudinal changes in ACC activity were further supported by a recent study investigating the neural effects of smartphone restriction using a CR task (Schmitgen et al., 2025). After a three-day restriction period, the most prominent time-dependent alterations in neural activation were observed in the ACC and nucleus accumbens (Fig. 2b). Moreover, significant associations emerged with excitatory neurotransmitters, notably the dopaminergic and serotonergic systems. At the behavioral level, craving intensity was positively associated with parietal activation (Schmitgen et al., 2025). Collectively, these findings suggest that ACC activity undergoes dynamic modulation in response to short-term smartphone restriction and therapeutic intervention, reflecting changes in neural circuits underlying reward processing and cognitive control (Seo et al., 2020; Schmitgen et al., 2025).

4. Discussion

Over the past decade, the body of neuroimaging research on PSU has grown steadily. Overall, the extent data delineates a largely convergent pattern across behavioral, structural, and functional domains, implicating neurocognitive networks subserving executive control, reward processing, affect regulation, and social cognition. Notably, these systems overlap significantly with neural circuits implicated in both substance-related and behavioral addictions (Zilverstand et al., 2018), further supporting the conceptualization of PSU within a broader addiction framework.

Considering the existing evidence, a striking observation is that, unlike the relatively well-explored domains of affective, attentional, and executive functioning, the neuroimaging literature on social cognitive processes in PSU remains notably limited. This is particularly surprising given the prominent (and potentially central) role of social cognition in PSU, as highlighted by recent research (Deng et al., 2021; Luque-Reca et al., 2024; Youn, 2019). Of particular relevance is the construct of FoMO, which has emerged as a robust psychological correlate of PSU (Zhang et al., 2023). FoMO is closely linked to the apprehension of social exclusion (Holte et al., 2022), thereby providing a direct experimental framework for probing social cognitive mechanisms in the context of PSU. To investigate the neural correlates of social exclusion in PSU, a recent fMRI study employed the Cyberball paradigm (Vijayakumar et al., 2017), i.e. a virtual ball-tossing task in which participants are first included in, and subsequently excluded from, a simulated social interaction (Henemann et al., 2025). This experimentally induced “social pain” has been validated as a reliable model for eliciting neural responses associated with experiences of rejection and isolation (Rotge et al., 2015). During the exclusion phase of the task, individuals with PSU exhibited significant hyperactivation in cingulate and superior frontal regions—areas implicated in the processing of social pain and the regulation of aversive affective states (Henemann et al., 2025) (Fig. 2c). In contrast, control participants demonstrated increased activation in parietal regions involved in the reappraisal of social cues and attentional orienting to external stimuli. These findings point to heightened sensitivity to social exclusion in individuals with PSU and underscore the potential role of FoMO and rejection-related anxiety in reinforcing problematic smartphone use behavior (Henemann et al., 2025).

The I-PACE model offers a theoretically robust framework for interpreting these findings. Within the I-PACE model, PSU can be understood as the outcome of dynamic interactions among individual predispositions, affective-motivational states, cognitive, and behavioral control mechanisms (Brand et al., 2019). This model provides a refined and integrative framework for systematically organizing the heterogeneous neurobiological evidence on PSU and for elucidating its functional relevance in the development and maintenance of maladaptive smartphone use:

1. At the level of *person-related* factors, individuals with PSU repeatedly exhibit structural and functional alterations in brain regions central to self-regulation, emotion, and reward processing, such as the insular cortex, OFC, and ACC (Martins et al., 2021; Morawetz and Basten, 2024). The extant neuroimaging evidence suggests a risk signature within frontolimbic networks, which can be interpreted as correlates of diminished executive control and self-regulatory capacity. Additional subcortical changes, including reduced structural integrity of the striatum and alterations in glutamatergic signaling pathways associated with reward sensitivity, further underscore a neural vulnerability toward

repetitive, yet eventually excessive, stimulus seeking and compensatory usage behaviors.

2. On the *affective-motivational* level, disturbances in stress regulation and reward expectation are particularly salient. Studies on INA report reduced activation in the ACC and insular cortex, together functional dysconnectivity within frontostriatal reward circuits, findings that also correlate with craving and withdrawal. Consistent with these observations are CR data, especially regarding the neural effects of smartphone restriction (Schmitgen et al., 2020; Henemann et al., 2023b; Schmitgen et al., 2025). The reward value inherent in CR likely involves a significant social component, as individuals with PSU exhibit enhanced neural responsiveness to social rewards (Deng et al., 2021). Conversely, social exclusion elicits increased activation in salience and emotion-related brain regions, indicating a heightened sensitivity to aversive social scenarios (Henemann et al., 2025).

3. At the *cognitive process* level, alterations in selective attention and executive functions appear particularly relevant. Task-based fMRI studies demonstrate reduced recruitment of prefrontal control regions, especially under conditions of heightened cognitive load (Asao et al., 2024). Multimodal data fusion methods further reveal diminished coupling between frontoparietal networks and executive control systems (Henemann et al., 2023a). Concurrently, INA studies in PSU indicate reduced functional flexibility in attention-related brain areas (Liu et al., 2022). These neurocognitive changes may represent a central mechanism underlying repetitive smartphone use patterns that – in the long run – can be difficult to interrupt.

4. At the *executive* level, neurobiological correlates of maladaptive decision-making and reinforcement mechanisms also emerge. CR activity patterns within well-known reward-related areas, such as the ventral striatum, OFC, and ACC, can predict individual PSU severity with high accuracy (Oweda et al., 2025). Neurochemical data on CR implicate the endogenous opioid system, particularly the μ-opioid receptor, in the processing of smartphone-specific cues (Henemann et al., 2023a). Additionally, short-term smartphone restriction is associated with alterations in dopaminergic and serotonergic neurotransmission, linked to craving and heightened CR (Schmitgen et al., 2025). Together, these findings support the hypothesis that reward and affective systems not only contribute to the initial development of PSU but also critically sustain it via conditioned reinforcement processes.

Within the I-PACE model, social cognitive processes may assume a particularly pivotal role. Smartphones have evolved into almost indispensable devices for maintaining social interaction and communication. However, the mere social attractiveness of these devices is unlikely to sufficiently account for the development of PSU. Rather, an affective-aversive component arising in social interaction contexts warrants consideration as a critical factor. In this regard, the concept of FoMO could offer a valuable extension of the I-PACE model not primarily as an affective phenomenon, but rather as a social cognitive distortion characterized by the expectation of being excluded from significant social information, which in turn leads to heightened attentional allocation toward digital communication channels (Zhang et al., 2023; Elhai et al., 2025). Within the context of PSU, FoMO can be understood as a motivational mechanism that maladaptively enhances social comparison processes and reward expectation (Wang et al., 2023; Stirnberg et al., 2024). Current multimodal neuroimaging findings in PSU substantiate this notion, providing evidence for structural associations between FoMO and self-referential processing (Wang et al., 2023), as well as altered neural responses to experiences of social exclusion (Henemann et al., 2025). We propose that the reinforcement mechanisms central to the I-PACE framework can be enhanced by recognizing that socially embedded rewards may constitute particularly potent drivers of PSU, given that smartphone use fundamentally occurs within social contexts (Montag et al., 2019b). By integrating social cognitive processes into the I-PACE model, we can better account for how digital social feedback loops amplify reward expectation and sustain cue reactivity patterns, thereby providing a more comprehensive understanding of PSU etiology and maintenance. Further refinement of the I-PACE model will be the focus of future research, particularly studies explicitly examining the interaction between smartphone-related CR and social cognitive processes. Notably, CR-associated neural effects, as demonstrated consistently in both cross-sectional and longitudinal designs (Schmitgen et al., 2020; Henemann et al., 2023b; Oweda et al., 2025; Schmitgen et al., 2025), exhibit a marked neuroanatomical overlap with networks engaged during social cognitive processes (Vijayakumar et al., 2017; Rotge et al., 2015). It is thus plausible (and empirically testable), that smartphone-related CR differs depending on whether the cues are embedded within social versus non-social contexts. In such cases, CR would be sustained not by the device per se, but by its social significance, i.e. a dynamic whose importance for the emergence and persistence of PSU cannot be overstated.

In this context, cultural variations in sensitivity to social rewards may differentially influence PSU through distinct pathways related to e.g. collectivistic versus individualistic orientations. Research demonstrates that individuals from collectivistic cultures, which emphasize interpersonal harmony and group cohesion, exhibit heightened neural sensitivity to social exclusion and enhanced activation of error-monitoring systems when errors occur in social contexts (Rapp et al., 2021). This heightened social vigilance may amplify the motivational salience of smartphone-mediated social feedback, particularly for maintaining group connectivity and avoiding social exclusion. It seems thinkable that individuals that live in collectivistic societies experience heightened FoMo and resulting drive to use the smartphone more frequently. Further, cross-cultural neuroimaging studies suggest that social information may be processed with greater cognitive and emotional weight in populations with collectivistic orientations (Han and Ma, 2015; Knyazev et al., 2018). Consequently, FoMO and related social cognitive distortions may manifest more intensely in collectivistic cultures, where smartphone use serves not merely as personal communication but as a critical tool for maintaining social bonds. While individualistic cultures may emphasize personal achievement and autonomy through smartphone use, collectivistic cultures prioritize relational maintenance and social approval that influence the neural and behavioral pathways toward PSU (Medina et al., 2019; Chiao et al., 2009). This cultural amplification of social reward sensitivity may explain differential PSU prevalence rates observed across cultures, with collectivistic societies potentially showing greater vulnerability to socially mediated smartphone addiction patterns.

We were fully aware of the potential limitations inherent in relying on a single database and had considered this issue prior to review registration. Our choice of PubMed was motivated by the aim to reduce heterogeneity in a field already marked by diverse conceptualizations, instruments, and methodologies (Panova and Carbonell, 2018; Montag et al., 2019a; James et al., 2023), variability that cannot be comprehensively addressed within the focused scope of this review. We acknowledge that PSU is a dynamic construct spanning multiple disciplines well beyond neuroscience, with ongoing controversies over definitions (Wu et al., 2021), psychometric measures (Harris et al., 2020), and the balance between detrimental and potentially beneficial effects (Wan et al., 2025). While deeper engagement with these broader debates would be valuable, doing so would exceed the boundaries of the present work. To further limit heterogeneity, we restricted inclusion to studies explicitly referencing "smartphones," thereby distinguishing PSU from earlier conceptualizations of mobile- or cell-phone use, which did not encompass the multifunctionality and app-driven ecosystem of modern devices (Harris et al., 2020; Rozgonjuk et al., 2016). Notably, the vast majority of the studies included in this review employed either SAS-SV or SAPS, reflecting some convergence in measurement despite continued diversity across instruments. Furthermore, previous reviews underscore the variability in both the number and scope of publications identified, even when employing multiple databases and comparable criteria (Anbumalar and Binu Sahayam, 2024; Leon Mendez et al., 2024; Li and Yang, 2024; Montag and Becker, 2023), suggesting that

additional database expansion would not necessarily yield substantially greater coverage. Our focused approach nonetheless yielded 35 eligible studies, offering sufficient material to investigate the neural mechanisms of PSU and to position these findings within the extended social-cognitive framework that guided our review. Still, it is important to note that our review encompasses PubMed-indexed studies published between January 2015 and April 2025. Accordingly, work appearing thereafter was not considered (Tie et al., 2025; Yao et al., 2025). Nonetheless, these most recent findings further reinforce the notion that PSU is associated with disrupted connectivity across networks mediating executive control and the regulation stressful contexts, in line with maladaptive neural alterations observed in other problematic and potentially addictive behaviors, including PSU.

5. Conclusion

To date, the extant findings in PSU consistently implicate neural systems that are central to executive control, reward processing, affective regulation, and social cognition. Collectively, these networks support socially attuned and attentionally/executively regulated processes that appear particularly susceptible to the converging reward and social feedback cues characteristic of smartphone use. As observed in both substance-use and behavioral addictions, repeated activation of these systems may lead to maladaptive behavioral and neural changes. These include an increased attentional bias toward smartphone-related stimuli, heightened valuation of socially embedded (yet predominantly digitally mediated) rewards, reduced top-down regulatory efficiency under inter- and intrapersonal stress conditions, and enhanced sensitivity to social threat or exclusion in offline contexts. From both a preventive and therapeutic perspective, these findings suggest that, in addition to targeting e.g. craving and impulsivity (Mac, 2020; Rahmillah et al., 2023), promoting cognitive restructuring of digital feedback, reorganizing social networks, and enhancing self-efficacy and self-esteem in real-world social interactions may constitute valuable extensions to existing preventive approaches and therapeutic strategies in clinically relevant cases.

From a methodological perspective, this review also notes a marked imbalance emerges between structural imaging studies and those investigating INA, as compared to paradigm-based imaging, which so far have for the most part focused on cognitive processes to a greater extent than on affective or emotional processing or social cognition. This discrepancy highlights an important avenue for future research, with a stronger emphasis on task-based imaging to further investigate affective and social-cognitive dimensions of PSU, ideally complemented by fusing structural and resting-state approaches that can clarify the complex interplay between structural variation, INA, and diverse domains of cognitive and affective processing (Henemann et al., 2023a; Schmitgen et al., 2022). Another clear need for future research is longitudinal research on PSU. This is essential as it enables the examination of intra-individual fluctuations in use intensity over time (Wazulin et al., 2025), thereby overcoming the limitations of cross-sectional designs. This methodological approach allows for the identification of distinct developmental trajectories of PSU and their psychosocial predictors, including self-regulation, FoMO, depressive symptoms, and anxiety, while also elucidating protective factors across critical transitions from adolescence to young adulthood (Xiao et al., 2025). Moreover, longitudinal data facilitate the analysis of gender-specific patterns and app-category preferences, which may remain undetected in short-term assessments (Wazulin et al., 2025). Ultimately, such designs are essential for assessing neural mechanisms as well as the effectiveness and long-term durability of targeted interventions in PSU (Seo et al., 2020; Schmitgen et al., 2025).

CRediT authorship contribution statement

Nadine D. Wolf: Writing – review & editing, Writing – original draft, Validation, Data curation, Conceptualization. **Gudrun M. Henemann:** Writing – review & editing, Validation, Supervision. **Mike M. Schmitgen:** Writing – review & editing, Visualization, Software, Resources. **Julian Koenig:** Writing – review & editing, Validation, Methodology. **Patrick Bach:** Writing – review & editing, Supervision, Methodology. **Robert C. Wolf:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.pnpbp.2025.111531.

Data availability

Data will be made available on request.

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